

For both education groups (with/without HS diploma), we estimate the fraction of the total county population (ages 25 years and above) in each education group. Thus, the two estimates sum to one for each county.

For each estimate, we generate a coefficient of variation (CV) equal to the ratio of the standard error to the point estimate. For counties with a CV greater than 0.3 (1.7% percent of all counties), we impute the county-level estimate with a state-level estimate following Census guidance, which defines any estimate with a CV greater than 0.3 as low reliability and to be used with extreme caution (King et al. 2015).

J.3.2 Poverty Status

To determine the poverty status at the county level, we utilize ACS 5-year estimates from 2018 to 2022. The resulting datasets represent the fraction of the total population in the county that falls below the federal poverty line (termed “below_poverty_line”) and the fraction of the population that falls above the poverty line (termed “above_poverty_line”). BenMAP-CE also includes two variables (termed “below_2x_poverty_line” and “above_2x_poverty_line”) representing the fraction of the county-level population below and above 200% of the poverty line. The county-level proportions below the poverty line range from 3-56% and below 200% of the poverty line range from 6-77%.

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J.3.3 Health Insurance

We use data from the Small Area Health Insurance Estimates (SAHIE) collected by the U.S. Census Bureau from 2018 to 2022 to calculate the percentage of individuals with and without health insurance in each county. The SAHIE data provides the number of individuals with and without health insurance by county, disaggregated into three age groups: 18 to 65, 40 to 65, and under 65. Using these data, we calculated the percent of people uninsured by county for the age groups 1 to 17, 18-39, 40-64, and under 65 (termed “pct_uninsured1_17”, “pct_uninsured18_39”, “pct_uninsured40_64”, and “pct_uninsuredUnder_65”). As an example of this calculation, we subtracted the 18 to 65 uninsured population from the under 65 uninsured population to generate the 1 to 17 uninsured population. This value is then divided by the total 1 to 17 population (derived in the same manner) to estimate the uninsured rate.

J.3.4 Blue Collar Workers

We use five-year estimates (2018-2022) from the ACS to estimate the fraction of each county’s labor force employed in white collar and blue collar occupations. The dataset includes the number of employed individuals over 16 that work within five occupation

categories. We assign each of these five occupations to either the blue collar or white collar designation, as shown in Table J-8.

Table J-8. Mapping Occupations to Blue Collar and White Collar Designations

Occupation	Designation
Management, business, science and arts	White collar
Service	White collar
Sales and office	White collar
Natural resources, construction and maintenance	Blue collar
Production, transportation and material moving	Blue collar

We calculate the fraction of each county in blue collar professions by dividing the total number of individuals employed in blue collar jobs by the total number of employed individuals within each county (termed “pct_BlueCollar”). The same calculation is done for white collar professions.

J.3.5 Linguistic Isolation

We use 5-year ACS estimates from 2018-2022 to estimate linguistic isolation at the county level. The resulting datasets represent the fraction of the total population age 5 and over in the county that are linguistically isolated (defined by speaking English “less than very well”, termed “english_less_than_verywell”) and the fraction of the population age 5 and over that speak English “very well or better” (termed “english_verywell_or_better”). This definition follows as closely as possible the definition of linguistic isolation used in other regulatory analyses. Additionally, we produced an alternative definition in which the linguistically isolated population is more restrictive, i.e., those who speak English less than “well” (termed “english_less_than_well”), and inversely, those who speak English “well or better” (termed “english_well_or_better”).

Instead of imputing in cases of uncertainty, we generate county-level point estimates of linguistic isolation in an effort to prioritize geographic specificity in these estimates.

Appendix K. Uncertainty & Pooling

This Appendix discusses the treatment of uncertainty in BenMAP, both for incidence changes and associated dollar benefits. Some background is then given on pooling methodology. Finally, the mechanics of the various Pooling Methods available in BenMAP are discussed in detail, including Subjective Weight based pooling, Fixed Effects pooling, Random / Fixed Effects pooling, and independent and dependent Sum and Subtraction.

K.1 Uncertainty

Although there are several sources of uncertainty affecting estimates of incidence changes and associated benefits, the sources of uncertainty that are most readily quantifiable in benefit analyses are uncertainty surrounding the health impact functions and uncertainty surrounding unit dollar values. The total dollar benefit associated with a given endpoint group depends on how much the endpoint group will change in the control scenario (e.g., how many premature deaths will be avoided) and how much each unit of change is worth (e.g., how much a statistical death avoided is worth).

Both the uncertainty about the incidence changes and uncertainty about unit dollar values can be characterized by distributions. Each “uncertainty distribution” characterizes our beliefs about what the true value of an unknown (e.g., the true change in incidence of a given health effect) is likely to be, based on the available information from relevant studies. Although such an “uncertainty distribution” is not formally a Bayesian posterior distribution, it is very similar in concept and function (see, for example, the discussion of the Bayesian approach in Kennedy 1990, pp. 168-172). Unlike a sampling distribution (which describes the possible values that an estimator of an unknown value might take on), this uncertainty distribution describes our beliefs about what values the unknown value itself might be.

Such uncertainty distributions can be constructed for each underlying unknown (such as a particular pollutant coefficient for a particular location) or for a function of several underlying unknowns (such as the total dollar benefit of a regulation). In either case, an uncertainty distribution is a characterization of our beliefs about what the unknown (or the function of unknowns) is likely to be, based on all the available relevant information. Uncertainty statements based on such distributions are typically expressed as 90 percent credible intervals. This is the interval from the fifth percentile point of the uncertainty distribution to the ninety-fifth percentile point. The 90 percent credible interval is a “credible range” within which, according to the available information (embodied in the uncertainty distribution of possible values), we believe the true value to lie with 90 percent probability. The uncertainty surrounding both incidence estimates and dollar benefits estimates can be characterized quantitatively in BenMAP. Each is described separately below.

K.1.1 Characterization of Uncertainty Surrounding Incidence Changes

To calculate point estimates of the changes in incidence of a given adverse health effect associated with a given set of air quality changes, BenMAP performs a series of calculations at each grid-cell. First, it accesses the health impact functions needed for the analysis, and then it accesses any data needed by the health impact functions. Typically, these include the grid-cell population, the change in population exposure at the grid-cell, and the appropriate baseline incidence rate. BenMAP then calculates the change in incidence of adverse health effects for each selected health impact function. The resulting incidence change is stored, and BenMAP proceeds to the next grid-cell, where the above process is repeated.

In *Latin Hypercube* mode, BenMAP reflects the uncertainty surrounding estimated incidence changes (resulting from the sampling uncertainty surrounding the pollutant coefficients in the health impact functions used) by producing a *distribution* of possible incidence changes rather than a single point estimate. To do this, it uses the distribution (*Dist Beta*) associated with the pollutant coefficient (*Beta*, or β), and potentially the point estimate (*Beta*) and two parameters (*P1Beta*, *P2Beta*). Typically, pollutant coefficients are normally distributed, with mean *Beta* and standard deviation *P1Beta*.

BenMAP uses an N-point Latin Hypercube to represent the underlying distribution of β and to create a corresponding distribution of incidence changes in each population grid cell, where N is specified by you. The Latin Hypercube method represents an underlying distribution by N percentile points of the distribution, where the *n*th percentile point is equal to:

$$(n - 1) \times \frac{100}{N} + \frac{100}{2N}$$

The Latin Hypercube method is used to enhance computer processing efficiency. It is a sampling method that divides a probability distribution into intervals of equal probability, with an assumption value for each interval assigned according to the interval's probability distribution. Compared with conventional Monte Carlo sampling, the Latin Hypercube approach is more precise over a fewer number of trials because the distribution is sampled in a more even, consistent manner (Decisioneering, 1996, pp. 104-105).

Suppose, for example, that you elect to use a 20-point Latin Hypercube. BenMAP would then represent the distribution of β by 20 percentile points, specifically the 2.5th, 7.5th, ..., 97.5th. To do this, the inverse cumulative distribution function specified by the distribution of β is called with the input probability equal to each of the 20 percentile points. BenMAP then generates an estimate of the incidence change in a grid-cell for each of these values of β , resulting in a distribution of N incidence changes. This distribution is stored, and BenMAP proceeds to the next population grid-cell, where the process is repeated.